Universidad de los Andes

Cálculo Integral con Ecuaciones (202319)

Ejercicios para practicar

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11.4 Criterio de comparación

(1) Determine si la serie converge o no. Procure usar el método de comparación directa o comparación por límite.

(a) $\displaystyle\sum_{n=0}^{\infty} \frac{n}{2n^3+1}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n}}$

(c) $\displaystyle\sum_{n=0}^{\infty} \frac{4+3^n}{2^n}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2-1}{3n^4+1}$

(e) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{1+2^n}$

(f) $\displaystyle\sum_{n=0}^{\infty} \frac{n}{(n+1)3^n}$

(g) $\displaystyle\sum_{n=0}^{\infty} \frac{\sqrt{n+1}}{2n^2+n+1}$

(h) $\displaystyle\sum_{n=2}^{\infty} \frac{\sin^2 n}{n^3}$

(i) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{\sqrt{4n^4+1}}$

(j) $\displaystyle\sum_{n=1}^{\infty} \frac{n-1}{n^2\sqrt{n}}$

(k) $\displaystyle\sum_{n=3}^{\infty} \frac{\cos^2 n}{n^3+1}$

(l) $\displaystyle\sum_{n=2}^{\infty} \frac{n+2}{(n+1)^2}$

(m) $\displaystyle\sum_{n=4}^{\infty} \frac{1}{n^{1+1/n}}$

(n) $\displaystyle\sum_{n=3}^{\infty} \frac{\mathrm{e}^{1/n}}{n}$

(ñ) $\displaystyle\sum_{n=2}^{\infty} \left(1+\frac{1}{n}\right)^2 \mathrm{e}^{-n}$

(o) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$